Graph Theory Results and Proofs

Jun Kil

Spring 2026

Contents

1 Degrees and Counting

1.1 — Degree-Sum Formula (Handshaking Lemma)

Theorem 1.1 (Degree-Sum Formula)**.** *For any graph G,*

$$\sum_{v \in V} d(v) = 2m.$$

Intuition. Every edge has two endpoints. When we sum up degrees, each edge gets counted exactly twice (once from each end). So the total degree must be twice the number of edges.

Proof Sketch. Use the incidence matrix M. Row sums give degrees; column sums are all 2 (each edge has two ends). So summing the whole matrix in two ways gives the result.

Step-by-step proof.

1. Let M be the incidence matrix of G, where rows correspond to vertices and columns to edges. Entry $M_{v,e} = 1$ if v is an endpoint of e, and 0 otherwise (with loops contributing 2 to their vertex).

2. The sum of entries in the row of vertex v equals $d(v)$, since it counts how many edges are incident to v.

3. Therefore, the total sum of all entries in M equals $\sum_{v \in V} d(v)$.

4. Each column of M (corresponding to an edge e) sums to 2, because every edge has exactly two endpoints.

5. Therefore, the total sum of all entries in M also equals $2m$.

6. Combining steps 3 and 5: $\sum_{v \in V} d(v) = 2m$.

Proof to memorize. Consider the incidence matrix M of G. The row sum for vertex v is $d(v)$, so the sum of all entries equals $\sum_{v \in V} d(v)$. But each column sum is 2 (each edge has two ends), so the sum of all entries is also $2m$. Hence $\sum_{v \in V} d(v) = 2m$. □

1.2 — Even Number of Odd-Degree Vertices

Corollary 1.2. *In any graph, the number of vertices of odd degree is even.*

Intuition. Since the sum of all degrees is $2m$ (even), and even-degree vertices contribute $0 \pmod 2$, the odd-degree vertices must together contribute an even amount. This means there must be an even number of them.

Proof Sketch. Take $\sum d(v) = 2m$ modulo 2. The left side reduces to the count of odd-degree vertices (mod 2), and the right side is 0. So the count is even.

Step-by-step proof.

1. By Theorem 1.1, $\sum_{v\in V} d(v) = 2m$.
2. Reduce modulo 2: $\sum_{v\in V} d(v) \equiv 0 \pmod 2$.
3. Each vertex with even degree contributes $0 \pmod 2$ to the sum.
4. Each vertex with odd degree contributes $1 \pmod 2$ to the sum.
5. So the left side is congruent to the number of odd-degree vertices modulo 2.
6. Therefore, the number of odd-degree vertices $\equiv 0 \pmod 2$, i.e., it is even.

Proof to memorize. By Theorem 1.1, $\sum_{v\in V} d(v) = 2m \equiv 0 \pmod 2$. Vertices of even degree contribute 0 to this sum mod 2, while vertices of odd degree each contribute 1. So the number of odd-degree vertices is congruent to 0 modulo 2, hence even. □

1.3 — Degree Inequality in Bipartite Graphs

Proposition 1.3. *Let $G[X,Y]$ be a bipartite graph without isolated vertices such that $d(x) \geq d(y)$ for all $xy \in E$, where $x \in X$ and $y \in Y$. Then $|X| \leq |Y|$, with equality if and only if $d(x) = d(y)$ for all $xy \in E$.*

Intuition. In a bipartite graph, the total number of edges can be counted from either side: $\sum_{x\in X} d(x) = |E| = \sum_{y\in Y} d(y)$. If each x has degree at least as large as its neighbours y, then X vertices are "high-degree" and Y vertices are "low-degree." To push the same number of edges through fewer high-degree vertices requires $|X| \leq |Y|$.

Proof Sketch. Construct a matrix from the bipartite adjacency matrix B by dividing each row x by $d(x)$. Row sums become 1, column sums become ≤ 1 (using $d(x) \geq d(y)$). Since all row sums are 1, the total matrix sum is $|X|$. Since all column sums are ≤ 1, the total is $\leq |Y|$. Hence $|X| \leq |Y|$.

Step-by-step proof.

1. Let B be the bipartite adjacency matrix of $G[X,Y]$ (rows indexed by X, columns by Y).
2. Construct B' by dividing row x by $d(x)$ for each $x \in X$ (this is valid since there are no isolated vertices, so $d(x) \geq 1$).
3. Each row sum of B' equals $d(x)/d(x) = 1$, so $\sum_{x,y} B'_{x,y} = |X|$.
4. For each $y \in Y$, the column sum of B' is $\displaystyle\sum_{\substack{x\in X\\ xy\in E}} \frac{1}{d(x)}$. Since $d(x) \geq d(y)$ for each edge xy, we get $\frac{1}{d(x)} \leq \frac{1}{d(y)}$. There are $d(y)$ terms in this sum, so:
$$\sum_{\substack{x\in X\\ xy\in E}} \frac{1}{d(x)} \leq \sum_{\substack{x\in X\\ xy\in E}} \frac{1}{d(y)} = \frac{d(y)}{d(y)} = 1.$$
5. Therefore $|X| = \sum_{x,y} B'_{x,y} \leq |Y|$.
6. If $|X| = |Y|$, then every column sum must equal exactly 1, meaning $\frac{1}{d(x)} = \frac{1}{d(y)}$ (i.e., $d(x) = d(y)$) for all $xy \in E$.

Proof to memorize. Since there are no isolated vertices, $d(x) \geq 1$ for all $x \in X$. For each $x \in X$, we can write $1 = \sum_{\substack{y\in Y\\ xy\in E}} \frac{1}{d(x)}$ (summing $\frac{1}{d(x)}$ over the $d(x)$ neighbours of x). Therefore:

$$|X| = \sum_{x\in X} 1 = \sum_{x\in X} \sum_{\substack{y\in Y\\ xy\in E}} \frac{1}{d(x)} \leq \sum_{y\in Y} \sum_{\substack{x\in X\\ xy\in E}} \frac{1}{d(y)} = \sum_{y\in Y} \frac{d(y)}{d(y)} = |Y|.$$

The inequality replaces $\frac{1}{d(x)}$ by $\frac{1}{d(y)}$ for each edge xy, which is valid since $d(x) \geq d(y)$ gives $\frac{1}{d(x)} \leq \frac{1}{d(y)}$. The second-to-last step uses the fact that vertex y has exactly $d(y)$ neighbours in X, each contributing $\frac{1}{d(y)}$ to the inner sum. If $|X| = |Y|$, the inequality must be an equality, forcing $d(x) = d(y)$ for all $xy \in E$. □

2 Subgraphs, Decompositions, and Edge Cuts

2.1 — Minimum Degree Two Implies a Cycle

Theorem 2.1. *Let G be a graph in which all vertices have degree at least two. Then G contains a cycle.*

Intuition. If you walk along a longest path and reach the last vertex, that vertex must have another neighbour (degree ≥ 2). That neighbour can't extend the path (it's already longest), so it must be a vertex already on the path, creating a cycle.

Proof Sketch. Reduce to simple graphs (loops and parallel edges immediately give cycles). Take a longest path $P = v_0 v_1 \cdots v_k$. The vertex v_k has a neighbour $v \neq v_{k-1}$. If $v \notin V(P)$, we can extend P, contradicting maximality. So $v = v_i$ for some i, and $v_i v_{i+1} \cdots v_k v_i$ is a cycle.

Step-by-step proof.

1. If G has a loop, that is a cycle. If G has parallel edges, they form a cycle. So assume G is simple.
2. Let $P = v_0 v_1 \cdots v_k$ be a longest path in G.
3. Since $d(v_k) \geq 2$, vertex v_k has a neighbour $v \neq v_{k-1}$.
4. If $v \notin V(P)$, then $v_0 v_1 \cdots v_k v$ is a longer path, contradicting the choice of P.
5. So $v = v_i$ for some $0 \leq i \leq k-2$.
6. Then $v_i v_{i+1} \cdots v_k v_i$ is a cycle in G.

Proof to memorize. If G has a loop or parallel edges, it trivially contains a cycle. So assume G is simple. Let $P = v_0 v_1 \cdots v_k$ be a longest path in G. Since $d(v_k) \geq 2$, it has a neighbour $v \neq v_{k-1}$. If $v \notin V(P)$, then P can be extended, a contradiction. So $v = v_i$ for some $i \leq k-2$, and $v_i v_{i+1} \cdots v_k v_i$ is a cycle. □

2.2 — Condition for Containing a Quadrilateral

Theorem 2.2. *Any simple graph G with* $\sum_{v \in V} \binom{d(v)}{2} > \binom{n}{2}$ *contains a quadrilateral (a C_4).*

Intuition. $\binom{d(v)}{2}$ counts paths of length 2 centered at v. Each such path is determined by its pair of endpoints. There are only $\binom{n}{2}$ possible endpoint pairs. If the total number of 2-paths exceeds $\binom{n}{2}$, by pigeonhole, two 2-paths share endpoints, and their union is a C_4.

Proof Sketch. Count paths of length 2. There are $p_2 = \sum_v \binom{d(v)}{2}$ such paths. Each 2-path has a unique pair of end-vertices, and there are only $\binom{n}{2}$ possible pairs. By Pigeonhole, if $p_2 > \binom{n}{2}$, two 2-paths share both endpoints, forming a C_4.

Step-by-step proof.

1. For each vertex v, the number of paths of length 2 with v as centre is $\binom{d(v)}{2}$ (choose 2 neighbours of v).
2. Each path of length 2 has a unique central vertex, so $p_2 = \sum_{v \in V} \binom{d(v)}{2}$.
3. Each path of length 2 is also determined by its pair of endpoints. There are at most $\binom{n}{2}$ such pairs.
4. By hypothesis, $p_2 > \binom{n}{2}$.
5. By the Pigeonhole Principle, two distinct paths of length 2 share the same pair of endpoints. Say these are u-w_1-v and u-w_2-v.
6. Since G is simple and $w_1 \neq w_2$, the union uw_1vw_2u forms a C_4.

Proof to memorize. Let p_2 denote the number of paths of length two in G. Each such path has a unique central vertex, and the number centred at v is $\binom{d(v)}{2}$ (choosing 2 of its neighbours), so $p_2 = \sum_{v \in V} \binom{d(v)}{2}$. Each 2-path also has a unique pair of end-vertices, and there are at most $\binom{n}{2}$ possible endpoint pairs. By hypothesis $p_2 > \binom{n}{2}$, so by the Pigeonhole Principle, two distinct 2-paths share the same pair of endpoints, say u-w_1-v and u-w_2-v with $w_1 \neq w_2$ (the paths are distinct but share endpoints, so they must differ in their centres). Then u, w_1, v, w_2 are four distinct vertices and uw_1, w_1v, vw_2, w_2u are four edges forming a C_4. $\square$

2.3 — Rédei's Theorem

Theorem 2.3 (Rédei)**.** *Every tournament has a directed Hamilton path.*

Intuition. In a tournament, every pair of vertices has an arc between them. We prove by induction: remove one vertex, find a Hamilton path in the rest, then insert the removed vertex back in the right spot.

Proof Sketch. Induction on n. Remove a vertex v; by induction, $T - v$ has a directed Hamilton path $P = (v_1, \ldots, v_{n-1})$. Try to insert v: if $v \to v_1$, prepend v. If $v_{n-1} \to v$, append v. Otherwise, $v_1 \to v$ and $v \to v_{n-1}$, so there exists some index i where $v_i \to v$ but $v \to v_{i+1}$. Insert v between v_i and v_{i+1}.

Step-by-step proof.

1. **Base case:** A tournament on 1 vertex trivially has a directed Hamilton path.
2. **Inductive step:** Assume every tournament on $n - 1$ vertices has a directed Hamilton path. Let T be a tournament on $n \geq 2$ vertices. Pick any vertex v.
3. $T' = T - v$ is a tournament on $n - 1$ vertices. By induction, T' has a directed Hamilton path $P = (v_1, v_2, \ldots, v_{n-1})$.
4. **Case 1:** If (v, v_1) is an arc, then $(v, v_1, \ldots, v_{n-1})$ is a Hamilton path of T.
5. **Case 2:** If (v_{n-1}, v) is an arc, then $(v_1, \ldots, v_{n-1}, v)$ is a Hamilton path of T.
6. **Case 3:** Otherwise, (v_1, v) and (v, v_{n-1}) are arcs of T. Since v is adjacent to every vertex of P, and the arc direction "switches" from $v_i \to v$ to $v \to v_{i+1}$ at some point, there exists i with $1 \leq i < n - 1$ such that (v_i, v) and (v, v_{i+1}) are both arcs.
7. Then $(v_1, \ldots, v_i, v, v_{i+1}, \ldots, v_{n-1})$ is a directed Hamilton path of T.

Proof to memorize. By induction on n. The base case $n = 1$ is trivial. For $n \geq 2$, pick $v \in V(T)$. By induction, $T - v$ has a directed Hamilton path $P = (v_1, \ldots, v_{n-1})$. Since T is a tournament, for each v_i on P, either (v, v_i) or (v_i, v) is an arc.

If (v, v_1) is an arc, prepend v to get the Hamilton path $(v, v_1, \ldots, v_{n-1})$.

If (v_{n-1}, v) is an arc, append v to get $(v_1, \ldots, v_{n-1}, v)$.

Otherwise, (v_1, v) and (v, v_{n-1}) are both arcs. For each i, label v_i as "in" if (v_i, v) is an arc and "out" if (v, v_i) is an arc. Then v_1 is "in" and v_{n-1} is "out," so there exists some index i where v_i is "in" and v_{i+1} is "out," i.e., (v_i, v) and (v, v_{i+1}) are both arcs. Then $(v_1, \ldots, v_i, v, v_{i+1}, \ldots, v_{n-1})$ is a directed Hamilton path of T. □

2.4 — Spanning Bipartite Subgraph with Large Degree

Theorem 2.4. *Every loopless graph G contains a spanning bipartite subgraph F such that $d_F(v) \geq \frac{1}{2}d_G(v)$ for all $v \in V$.*

Intuition. Among all spanning bipartite subgraphs, pick one with the most edges. If any vertex v had fewer than half its edges crossing the bipartition, we could "move" v to the other side and gain more crossing edges, a contradiction.

Proof Sketch. Take a spanning bipartite subgraph $F[X, Y]$ with the maximum number of edges. Suppose some $v \in X$ has $d_F(v) < \frac{1}{2}d_G(v)$. Move v from X to Y: edges from v into Y are lost ($d_F(v)$ edges) but edges from v into X are gained ($d_G(v) - d_F(v)$ edges). Net change: $d_G(v) - 2d_F(v) > 0$, contradicting maximality of F.

Step-by-step proof.

1. G is loopless, so it has spanning bipartite subgraphs (e.g., the empty one).
2. Let $F = F[X, Y]$ be a spanning bipartite subgraph with the maximum number of edges.
3. Suppose for contradiction that $d_F(v) < \frac{1}{2}d_G(v)$ for some $v \in X$.
4. Construct F': the spanning bipartite subgraph with parts $X' = X \setminus \{v\}$ and $Y' = Y \cup \{v\}$, keeping all edges of G that cross this new bipartition.
5. Compared to F: edges from v to Y (there were $d_F(v)$ of them) are lost; edges from v to X (there are $d_G(v) - d_F(v)$ of them) are gained.
6. So $e(F') = e(F) - d_F(v) + (d_G(v) - d_F(v)) = e(F) + d_G(v) - 2d_F(v)$.
7. Since $d_F(v) < \frac{1}{2}d_G(v)$, we get $d_G(v) - 2d_F(v) > 0$, so $e(F') > e(F)$.
8. This contradicts the choice of F as having the maximum number of edges.

Proof to memorize. Since G is loopless, spanning bipartite subgraphs exist (e.g., the empty one). Among all spanning bipartite subgraphs of G, let $F = F[X, Y]$ be one with the most edges. Suppose for contradiction that $d_F(v) < \frac{1}{2}d_G(v)$ for some v; without loss of generality $v \in X$.

Move v from X to Y: define F' with bipartition $(X \setminus \{v\},\ Y \cup \{v\})$ containing all edges of G that cross this new partition. The edges of G incident to v split into two types: those going to Y (which were in F, and there are $d_F(v)$ of them) and those going to $X \setminus \{v\}$ (which were *not* in F, and there are $d_G(v) - d_F(v)$ of them). After moving v, the former are lost and the latter are gained. All other edges of F (not incident to v) remain, since they still cross the bipartition. Thus:

$$e(F') = e(F) - d_F(v) + (d_G(v) - d_F(v)) = e(F) + (d_G(v) - 2d_F(v)) > e(F),$$

since $d_F(v) < \frac{1}{2}d_G(v)$ implies $d_G(v) - 2d_F(v) > 0$. This contradicts the maximality of F. □

2.5 — Average Degree Implies Large Minimum Degree Subgraph

Theorem 2.5. *Every graph with average degree at least* $2k$*, where* k *is a positive integer, has an induced subgraph with minimum degree at least* $k+1$*.*

Intuition. A graph with high average degree must contain a "dense core." If we pick the densest induced subgraph, every vertex in it must have high degree, because removing a low-degree vertex would increase (or maintain) the average degree with fewer vertices, contradicting our choice.

Proof Sketch. Pick an induced subgraph F with the largest average degree and (subject to that) the fewest vertices. If some vertex v has $d_F(v) \le k$, removing v gives $F' = F - v$ with average degree $\ge d(F)$ and fewer vertices, contradicting the choice.

Step-by-step proof.

1. Let F be an induced subgraph of G with the maximum average degree $d(F)$, and subject to that, the smallest number of vertices.
2. Note $d(F) \ge d(G) \ge 2k$ by choice.
3. Suppose $d_F(v) \le k$ for some $v \in V(F)$. (We need $v(F) > 1$; if $v(F) = 1$, then $\delta(F) = d(F) \ge 2k \ge k+1$ and we are done.)
4. Let $F' = F - v$. Then:
$$d(F') = \frac{2e(F')}{v(F')} = \frac{2e(F) - 2d_F(v)}{v(F)-1} \ge \frac{2e(F) - 2k}{v(F)-1}.$$
5. Since $d(F) = 2e(F)/v(F)$, we have $2e(F) = d(F) \cdot v(F)$, so:
$$d(F') \ge \frac{d(F)\cdot v(F) - 2k}{v(F)-1} \ge \frac{d(F)\cdot v(F) - d(F)}{v(F)-1} = d(F).$$
(using $2k \le d(F)$).
6. So $d(F') \ge d(F)$ with $v(F') < v(F)$, contradicting the choice of F.
7. Therefore $d_F(v) \ge k+1$ for all v, i.e., $\delta(F) \ge k+1$.

Proof to memorize. Let F be an induced subgraph of G with the largest average degree and, subject to this, the fewest vertices. We have $d(F) \ge d(G) \ge 2k$.

If $v(F) = 1$ then F is a single vertex (possibly with loops), and $\delta(F) = d(F) \ge 2k \ge k+1$, so we are done.

Otherwise $v(F) > 1$. Suppose for contradiction that $d_F(v) \le k$ for some vertex v of F. Let $F' = F - v$, which is also an induced subgraph of G. Removing v removes exactly $d_F(v)$ edges, so $e(F') = e(F) - d_F(v)$. Thus:

$$d(F') = \frac{2e(F) - 2d_F(v)}{v(F)-1}.$$

Since $d(F) = 2e(F)/v(F)$, we have $2e(F) = d(F)\cdot v(F)$, and since $d_F(v) \le k \le d(F)/2$, we get $2d_F(v) \le d(F)$. Substituting:

$$d(F') = \frac{d(F)\cdot v(F) - 2d_F(v)}{v(F)-1} \ge \frac{d(F)\cdot v(F) - d(F)}{v(F)-1} = \frac{d(F)(v(F)-1)}{v(F)-1} = d(F).$$

So F' has average degree $\ge d(F)$ but fewer vertices than F, contradicting the choice of F. Therefore $d_F(v) \ge k+1$ for every vertex v, i.e., $\delta(F) \ge k+1$. $\square$

2.7 — Veblen's Theorem

Theorem 2.7 (Veblen)**.** *A graph admits a cycle decomposition if and only if it is even (every vertex has even degree).*

Intuition. ($\Leftarrow$) If every vertex has even degree, we can peel off cycles one at a time. After removing a cycle, all degrees are still even, so we can repeat until no edges remain.

($\Rightarrow$) If the graph decomposes into cycles, each cycle contributes 0 or 2 to each vertex's degree, so all degrees are even.

Proof Sketch. ($\Rightarrow$) Each cycle contributes exactly 2 to the degree of each vertex it passes through, and 0 to others. So degrees are even.

($\Leftarrow$) By induction on $e(G)$. If G is empty, the empty family works. Otherwise, the non-isolated vertices all have degree ≥ 2, so by Theorem 2.1, G contains a cycle C. Remove C; the result $G' = G \setminus E(C)$ is still even (subtracting 2 from each vertex on C), with fewer edges. By induction, G' has a cycle decomposition, and adding C gives one for G.

Step-by-step proof.

1. ($\Rightarrow$) If G has a cycle decomposition, each cycle contributes exactly 2 to the degree of each vertex on it. So all degrees are even.
2. ($\Leftarrow$) By induction on $e(G)$.
3. **Base case:** If $e(G) = 0$, the empty family is a cycle decomposition.
4. **Inductive step:** Suppose G is even and $e(G) > 0$.
5. Let F be the subgraph induced by the vertices of positive degree. Since G is even, F is even, so every vertex of F has degree ≥ 2.
6. By Theorem 2.1, F (and hence G) contains a cycle C.
7. Let $G' = G \setminus E(C)$. Removing a cycle subtracts 0 or 2 from each vertex's degree, so G' is even.
8. G' has fewer edges, so by induction, G' has a cycle decomposition $\mathcal{C}'$.
9. Then $\mathcal{C} = \mathcal{C}' \cup \{C\}$ is a cycle decomposition of G.

Proof to memorize. ($\Rightarrow$) If G has a cycle decomposition $\{C_1, \ldots, C_t\}$, then each edge belongs to exactly one cycle, and each cycle contributes exactly 2 to the degree of each vertex it passes through (entering and leaving), and 0 to vertices it does not pass through. So $d_G(v) = \sum_i d_{C_i}(v)$, where each $d_{C_i}(v) \in \{0, 2\}$. Hence all degrees are even.

($\Leftarrow$) By induction on $e(G)$. If $e(G) = 0$, the empty family is a (vacuous) cycle decomposition. If $e(G) > 0$, then since G is even, every vertex has even degree. Consider the subgraph F of G induced by the vertices of positive degree. Every vertex of F has even degree ≥ 2. By Theorem 2.1 (minimum degree ≥ 2 implies a cycle exists), F contains a cycle C. Let $G' = G \setminus E(C)$ (remove the edges of C). For each vertex v: if $v \notin V(C)$, then $d_{G'}(v) = d_G(v)$ (even); if $v \in V(C)$, then $d_{G'}(v) = d_G(v) - 2$ (still even, since we subtract 2). So G' is even, with $e(G') < e(G)$. By induction, G' has a cycle decomposition $\mathcal{C}'$. Then $\mathcal{C}' \cup \{C\}$ is a cycle decomposition of G. $\square$

2.8 — Decomposition of K_n into Complete Bipartite Graphs

Theorem 2.8. *Let $\mathcal{F} = \{F_1, F_2, \ldots, F_k\}$ be a decomposition of K_n into complete bipartite graphs. Then $k \geq n - 1$.*

Intuition. Think of this as assigning a numerical "weight" x_v to every node in the graph. We want to calculate a global metric—the sum of the products of the weights across every edge in K_n. We call this the **Edge Score**: $S = \sum_{uv \in E(K_n)} x_u x_v$.

The proof works by calculating this exact same Edge Score in two different ways:

1. **Globally (via K_n):** By forcing the sum of all node weights to be 0, the math dictates that the total Edge Score S must be strictly negative.
2. **Locally (via the decomposed F_i pieces):** By forcing the sum of the left-side weights of every bipartite piece to be 0, the Edge Score of every individual piece becomes 0. If every piece is 0, the total Edge Score S must be exactly zero.

A number cannot be both negative and zero. This contradiction happens because we assumed we had fewer than $n - 1$ pieces, which gave us the mathematical "degrees of freedom" (more variables than constraints) to assign these malicious, non-zero weights in the first place.

Proof Sketch. Let F_i have bipartition (X_i, Y_i). Set up a system of linear constraints on the node weights x_v:

1. The sum of all weights is zero: $\sum_{v \in V} x_v = 0$
2. The sum of the left-side weights for each bipartite graph is zero: $\sum_{v \in X_i} x_v = 0$

If $k < n - 1$, we have at most $n - 1$ constraints for n variables. By linear algebra, an overparameterized homogeneous system guarantees a nontrivial (non-zero) weight assignment. Calculate the total Edge Score twice. First, expanding $\left(\sum_v x_v\right)^2 = 0$ reveals the total Edge Score is negative. Second, adding up the edge scores of the individual F_i pieces yields zero, because $\left(\sum_{X_i} x_v\right)\left(\sum_{Y_i} x_v\right) = 0 \times \cdots = 0$. Negative cannot equal zero, hence $k \geq n - 1$.

Proof to memorize. Let $V = V(K_n)$ and let F_i have bipartition (X_i, Y_i) for $1 \leq i \leq k$. Consider a homogeneous system of $k + 1$ linear equations in n unknown weights $\{x_v : v \in V\}$:

$$\sum_{v \in V} x_v = 0, \qquad \sum_{v \in X_i} x_v = 0 \quad \text{for each } 1 \leq i \leq k.$$

Suppose for contradiction that $k < n - 1$. Then we have $k + 1 < n$ equations constraining n unknowns. By linear algebra, since there are fewer equations than variables, a nontrivial solution $(x_v)_{v \in V}$ exists (not all x_v are zero).

Claim 1: The total edge score of K_n is strictly negative.
From our first equation, $\sum_{v \in V} x_v = 0$. Therefore, the square of this sum is also zero. Expanding the square gives:

$$0 = \left(\sum_{v \in V} x_v\right)^2 = \sum_{v \in V} x_v^2 + 2 \sum_{uv \in E(K_n)} x_u x_v$$

Rearranging this to solve for the edge score:

$$\sum_{uv \in E(K_n)} x_u x_v = -\frac{1}{2} \sum_{v \in V} x_v^2$$

Because our solution is nontrivial, at least one $x_v \neq 0$, meaning the sum of squares $\sum x_v^2$ is strictly positive. Therefore, $-\frac{1}{2} \sum x_v^2 < 0$.

Claim 2: The total edge score of K_n is exactly zero.
Since $\mathcal{F}$ is a decomposition, the total edge score of K_n is the sum of the edge scores of its pieces F_i. Because F_i is a complete bipartite graph $K[X_i, Y_i]$, its edge score factors perfectly into the sum of its parts:

$$\sum_{uv \in E(F_i)} x_u x_v = \left(\sum_{u \in X_i} x_u \right) \left(\sum_{v \in Y_i} x_v \right)$$

By our system of equations, $\sum_{u \in X_i} x_u = 0$ for every piece. Therefore, the edge score of every F_i is 0. Summing them up yields the total edge score:

$$\sum_{uv \in E(K_n)} x_u x_v = \sum_{i=1}^{k} \sum_{uv \in E(F_i)} x_u x_v = \sum_{i=1}^{k} 0 = 0.$$

Contradiction: Claim 1 shows the sum $\sum_{uv \in E(K_n)} x_u x_v < 0$, but Claim 2 shows the exact same sum equals 0. This is impossible. Our assumption that $k < n-1$ must be false. Therefore, $k \geq n-1$. $\square$

2.9 — Edge Cut Formula

Theorem 2.9. *For any graph G and any subset X of V,*

$$|\partial(X)| = \sum_{v \in X} d(v) - 2e(X),$$

where $\partial(X)$ denotes the set of edges with exactly one end in X, and $e(X) = |E(G[X])|$.

Intuition. $\sum_{v \in X} d(v)$ counts every edge incident to a vertex in X. Edges with both ends in X get counted twice; edges with exactly one end in X (the cut edges) are counted once. Subtracting $2e(X)$ removes the double-counted internal edges, leaving exactly $|\partial(X)|$.

Proof Sketch. Count edges incident to X in two ways. The sum $\sum_{v \in X} d(v)$ counts each edge inside X twice and each cut edge once. So $\sum_{v \in X} d(v) = 2e(X) + |\partial(X)|$.

Step-by-step proof.

1. Each edge of G incident to a vertex in X is one of two types:
   - An edge with both ends in X (internal edge).
   - An edge with exactly one end in X (cut edge in $\partial(X)$).
2. $\sum_{v \in X} d(v)$ counts each internal edge twice (once from each endpoint in X) and each cut edge once (from its endpoint in X).
3. Therefore: $\sum_{v \in X} d(v) = 2e(X) + |\partial(X)|$.
4. Rearranging: $|\partial(X)| = \sum_{v \in X} d(v) - 2e(X)$.

Proof to memorize. When we sum $d(v)$ over all $v \in X$, each edge with both ends in X is counted twice, and each edge with exactly one end in X is counted once. That is, $\sum_{v \in X} d(v) = 2e(X) + |\partial(X)|$, whence $|\partial(X)| = \sum_{v \in X} d(v) - 2e(X)$. □

2.10 — Even Graphs and Edge Cuts

Theorem 2.10. *A graph G is even if and only if $|\partial(X)|$ is even for every subset X of V.*

Intuition. ($\Leftarrow$) If every edge cut has even size, in particular $|\partial(\{v\})|$ is even for each vertex, so each degree is even.

($\Rightarrow$) If all degrees are even, then by Theorem 2.9, $|\partial(X)| = \sum_{v \in X} d(v) - 2e(X)$ is even (a sum of even numbers minus an even number).

Proof Sketch. ($\Leftarrow$): $\partial(\{v\})$ is the set of links at v. If it is even, and loops contribute 2 to degree, then $d(v)$ is even.

($\Rightarrow$): Theorem 2.9 gives $|\partial(X)| = \sum_{v \in X} d(v) - 2e(X)$. If all degrees are even, both terms are even, so $|\partial(X)|$ is even.

Step-by-step proof.

1. ($\Leftarrow$) Suppose $|\partial(X)|$ is even for all $X \subseteq V$. Take $X = \{v\}$. Then $\partial(\{v\})$ is the set of links incident to v. Since loops contribute 2 to $d(v)$, and links contribute 1 each, we get $d(v) = |\partial(\{v\})| + 2 \cdot (\text{loops at } v)$, which is even.

2. ($\Rightarrow$) Suppose G is even (all degrees even). By Theorem 2.9:

$$|\partial(X)| = \sum_{v \in X} d(v) - 2e(X).$$

Each $d(v)$ is even, so $\sum_{v \in X} d(v)$ is even. Also $2e(X)$ is even. So $|\partial(X)|$ is even.

Proof to memorize. ($\Leftarrow$) If $|\partial(X)|$ is even for all X, take $X = \{v\}$. Then $|\partial(\{v\})|$ is the number of links at v, which is even. Since loops contribute 2 to degree, $d(v)$ is even.

($\Rightarrow$) If G is even, then by Theorem 2.9, $|\partial(X)| = \sum_{v \in X} d(v) - 2e(X)$, which is even since all $d(v)$ are even. $\square$

2.15 — Characterization of Bonds

Theorem 2.15. *In a connected graph G, a nonempty edge cut $\partial(X)$ is a bond if and only if both $G[X]$ and $G[V \setminus X]$ are connected.*

Intuition. A bond is a "minimal" border wall. If the territory on either side is broken into disconnected islands, you could draw a smaller border around just one island, meaning the original wall wasn't minimal. Conversely, if both sides are solid (connected) internally, leaving even a single gate (edge) open in the wall connects the whole world together. So, you must cut every single edge to separate them, making the cut minimal.

Proof Sketch. ($\Rightarrow$) If $\partial(X)$ is a bond and $G[X]$ is disconnected, say into components A and B, then the edges leaving A only go to $V \setminus X$. Thus $\partial(A) \subsetneq \partial(X)$, which contradicts minimality. ($\Leftarrow$) If both $G[X]$ and $G[V \setminus X]$ are connected, any subset of edges $S \subsetneq \partial(X)$ leaves at least one edge e uncut between them. Because both sides are internally connected, e acts as a bridge uniting the whole graph. So S cannot disconnect G, making $\partial(X)$ minimal.

Step-by-step proof.

1. ($\Rightarrow$) Suppose $\partial(X)$ is a bond. We want to show $G[X]$ is connected.
2. Suppose for contradiction $G[X]$ is disconnected. Let A be one connected component of $G[X]$, and B be the rest of X.
3. Since there are no edges between A and B, every edge incident to A must go directly to $V \setminus X$.
4. Therefore, the edge cut $\partial(A)$ consists only of edges that are also in $\partial(X)$. So, $\partial(A) \subseteq \partial(X)$.
5. Since G is connected and B is not empty, there are other edges in $\partial(X)$ attached to B. Thus, $\partial(A)$ is strictly smaller than $\partial(X)$, meaning $\partial(A) \subsetneq \partial(X)$.
6. This contradicts the fact that $\partial(X)$ is a bond (a minimal edge cut). Hence, $G[X]$ must be connected. By symmetry, $G[V \setminus X]$ is also connected.
7. ($\Leftarrow$) Suppose $G[X]$ and $G[V \setminus X]$ are both connected. We want to show $\partial(X)$ is a bond.
8. Suppose for contradiction $\partial(X)$ is not a bond. Then there is a strictly smaller edge cut $S \subsetneq \partial(X)$ that still disconnects the graph.
9. Since $S \subsetneq \partial(X)$, there is at least one edge $e = uv \in \partial(X)$ that is not in S. Let $u \in X$ and $v \in V \setminus X$.
10. Because $G[X]$ is connected, every vertex in X has a path to u. Because $G[V \setminus X]$ is connected, every vertex in $V \setminus X$ has a path to v.
11. Since e connects u and v, removing only S leaves a full path between any vertex in X and any vertex in $V \setminus X$.
12. This means removing S fails to disconnect the graph, contradicting that S is an edge cut. Thus, $\partial(X)$ is minimal and is a bond.

Proof to memorize. ($\Rightarrow$) Assume $\partial(X)$ is a bond (a minimal edge cut). Suppose for contradiction that $G[X]$ is disconnected. Then X can be partitioned into two non-empty sets, A and B, with no edges between them. Consider the edge cut $\partial(A)$. Since there are no edges between A and B, any edge leaving A must go to $V \setminus X$. Therefore, all edges in $\partial(A)$ are also in $\partial(X)$, meaning $\partial(A) \subseteq \partial(X)$. Furthermore, since G is connected, there must be edges from B to $V \setminus X$ as well, which means $\partial(A)$ is strictly smaller than $\partial(X)$ (i.e., $\partial(A) \subsetneq \partial(X)$). But $\partial(A)$ is a valid nonempty edge cut, which contradicts the minimality of $\partial(X)$. Thus, $G[X]$ must be connected. By symmetry, $G[V \setminus X]$ is also connected.

($\Leftarrow$) Assume $G[X]$ and $G[V \setminus X]$ are connected. We must show $\partial(X)$ is minimal. Suppose for contradiction there is a strictly smaller edge cut $S \subsetneq \partial(X)$. Because S is strictly smaller, there is at least one edge $e = uv$ in $\partial(X)$ that is not in S, where $u \in X$ and $v \in V \setminus X$. Since $G[X]$ is connected, every vertex in X can reach u using internal edges. Since $G[V \setminus X]$ is connected, every vertex in $V \setminus X$ can reach v using internal edges. The edge e connects u and v. Therefore, even if we remove the cut S, the graph remains completely connected through the intact bridge e. This contradicts the assumption that S is an edge cut. Therefore, no smaller edge cut exists, and $\partial(X)$ is a bond. $\square$

2.16 — Symmetric Difference of Even Subgraphs

Corollary 2.16. *The symmetric difference of two even subgraphs is an even subgraph.*

Intuition. An even subgraph is characterized by all its edge cuts being even (Theorem 2.10). The symmetric difference of two even-cardinality sets is even. So edge cuts of the symmetric difference, being the symmetric difference of two even edge cuts, are even.

Proof Sketch. For any $X \subseteq V$: $\partial_{F_1 \triangle F_2}(X) = \partial_{F_1}(X) \triangle \partial_{F_2}(X)$. Both $\partial_{F_1}(X)$ and $\partial_{F_2}(X)$ have even cardinality (by Theorem 2.10). The symmetric difference of two even-sized sets is even. So by Theorem 2.10, $F_1 \triangle F_2$ is even.

Step-by-step proof.

1. Let F_1 and F_2 be even subgraphs of G, and let $X \subseteq V$.
2. By Proposition 2.13 (edge cuts distribute over symmetric difference): $\partial_{F_1 \triangle F_2}(X) = \partial_{F_1}(X) \triangle \partial_{F_2}(X)$.
3. Since F_1 is even, $|\partial_{F_1}(X)|$ is even (by Theorem 2.10).
4. Since F_2 is even, $|\partial_{F_2}(X)|$ is even (by Theorem 2.10).
5. For any two sets A, B: $|A \triangle B| = |A| + |B| - 2|A \cap B|$. If $|A|$ and $|B|$ are both even, then $|A \triangle B|$ is even.
6. So $|\partial_{F_1 \triangle F_2}(X)|$ is even for every X.
7. By Theorem 2.10, $F_1 \triangle F_2$ is even.

Proof to memorize. Let F_1, F_2 be even subgraphs and $X \subseteq V$. By Proposition 2.13, $\partial_{F_1 \triangle F_2}(X) = \partial_{F_1}(X) \triangle \partial_{F_2}(X)$. By Theorem 2.10, both $|\partial_{F_1}(X)|$ and $|\partial_{F_2}(X)|$ are even. The symmetric difference of two even-cardinality sets is even (since $|A \triangle B| = |A| + |B| - 2|A \cap B|$). So $|\partial_{F_1 \triangle F_2}(X)|$ is even for all X, and by Theorem 2.10, $F_1 \triangle F_2$ is even. $\square$

2.18 — Even Subgraphs Meet Edge Cuts Evenly

Proposition 2.18. *In any graph, every even subgraph meets every edge cut in an even number of edges.*

Intuition. A cycle crosses into a set X exactly as many times as it crosses out, so it meets any edge cut an even number of times. Since every even subgraph admits a cycle decomposition (Theorem 2.7), the total intersection is a sum of even numbers, hence even.

Proof Sketch. Step 1: Show every cycle meets every edge cut evenly (as you traverse the cycle, each crossing into X is matched by a crossing out). Step 2: Every even subgraph admits a cycle decomposition (Theorem 2.7), so the count is a sum of even numbers.

Step-by-step proof.

1. **Claim:** Every cycle meets every edge cut in an even number of edges.
2. Let C be a cycle and $\partial(X)$ an edge cut. Each vertex of C is either in X or in $V \setminus X$.
3. As we traverse C, each time we move from X to $V \setminus X$ we cross an edge of $\partial(X)$, and each time we move from $V \setminus X$ to X we also cross an edge of $\partial(X)$.
4. Since C is a closed walk, the number of crossings from X to $V \setminus X$ equals the number from $V \setminus X$ to X.
5. Therefore $|E(C) \cap \partial(X)|$ is even.
6. Since F is an even subgraph (every vertex has even degree), by Theorem 2.7 (Veblen), F admits a cycle decomposition $\{C_1, C_2, \ldots, C_t\}$; that is, the edges of F can be partitioned into edge-disjoint cycles.
7. So $|E(F) \cap \partial(X)| = \sum_{i=1}^{t} |E(C_i) \cap \partial(X)|$, which is a sum of even numbers, hence even.

Proof to memorize. First, we show that any single cycle C meets an edge cut $\partial(X)$ in an even number of edges. As we traverse C, we may travel along many edges entirely within X or entirely within $V \setminus X$; these do not belong to $\partial(X)$. However, every time the cycle traverses an edge to *leave* X and enter $V \setminus X$, it must eventually traverse another edge to *return* to X in order to close the loop. Thus, the edges of C that cross the cut $\partial(X)$ always come in exit-return pairs. This means $|E(C) \cap \partial(X)|$ must be an even number.

Second, let F be an even subgraph. By definition, every vertex in F has an even degree. By Veblen's Theorem (Theorem 2.7), F admits a cycle decomposition; meaning its edges can be completely partitioned into a set of edge-disjoint cycles $\{C_1, C_2, \ldots, C_t\}$. The total number of edges F shares with the cut $\partial(X)$ is simply the sum of the crossings from each individual cycle:

$$|E(F) \cap \partial(X)| = \sum_{i=1}^{t} |E(C_i) \cap \partial(X)|$$

Since we established that each $|E(C_i) \cap \partial(X)|$ is an even number, their sum is also an even number. Therefore, F meets the edge cut evenly. □

2.20 — Kelly's Lemma

Lemma 2.20 (Kelly's Lemma)**.** *For any two graphs F and G such that $v(F) < v(G)$, the parameter $\binom{G}{F}$ is reconstructible.*

Proof. We count, in two ways, the total number of appearances of copies of F in the vertex-deleted subgraphs $G - v$.

Take a particular copy of F in G. This copy uses exactly $v(F)$ vertices of G, so there are

$$v(G) - v(F)$$

vertices of G that do *not* belong to it. If we delete one of those vertices, the copy remains present in $G - v$. On the other hand, if we delete a vertex of the copy itself, then that copy disappears.

Therefore each copy of F in G appears in exactly $v(G) - v(F)$ of the cards $G - v$.

Since there are $\binom{G}{F}$ copies of F in G, the total number of such appearances over all cards is

$$(v(G) - v(F))\binom{G}{F}. \tag{1}$$

Now count the same quantity in a different way. For each vertex $v \in V(G)$, the graph $G - v$ contains exactly $\binom{G-v}{F}$ copies of F. So the total number of appearances of copies of F in all cards is also

$$\sum_{v \in V(G)} \binom{G - v}{F}. \tag{2}$$

Equating (1) and (2), we get

$$\sum_{v \in V(G)} \binom{G - v}{F} = (v(G) - v(F))\binom{G}{F}.$$

Hence

$$\binom{G}{F} = \frac{1}{v(G) - v(F)} \sum_{v \in V(G)} \binom{G - v}{F}.$$

The right-hand side depends only on the deck of G, so $\binom{G}{F}$ is reconstructible. □

2.21 — Copies of F Through a Given Vertex

Corollary 2.21. *For any two graphs F and G such that $v(F) < v(G)$, the number of subgraphs of G that are isomorphic to F and include a given vertex v is reconstructible.*

Proof. The copies of F that contain v are exactly the copies of F in G minus the copies of F in $G - v$. Hence their number is

$$\binom{G}{F} - \binom{G-v}{F}.$$

By Kelly's Lemma (Lemma 2.20), $\binom{G}{F}$ is reconstructible, and $\binom{G-v}{F}$ is read directly from the card $G - v$. Therefore the required number is reconstructible. □

2.22 — Size and Degree Sequence are Reconstructible

Corollary 2.22. *The size and the degree sequence are reconstructible parameters.*

Proof. Take $F = K_2$. Then

$$e(G) = \binom{G}{K_2},$$

so the size is reconstructible by Kelly's Lemma (Lemma 2.20).

Also, for each vertex v, the number of copies of K_2 containing v is exactly $\deg(v)$. By Corollary 2.21 this number is reconstructible. Hence every degree is reconstructible, and therefore so is the degree sequence. □

2.25 — The Möbius Inversion Formula

Theorem 2.25 (The Möbius Inversion Formula)**.** *Let $f : 2^T \to \mathbb{R}$ be a real-valued function defined on the subsets of a finite set T. Define the function $g : 2^T \to \mathbb{R}$ by*

$$g(S) := \sum_{S \subseteq X \subseteq T} f(X). \tag{2.4}$$

Then, for all $S \subseteq T$,

$$f(S) = \sum_{S \subseteq X \subseteq T} (-1)^{|X|-|S|} g(X). \tag{2.5}$$

Proof. Substituting (2.4) into the right-hand side of (2.5),

$$\sum_{S \subseteq X \subseteq T} (-1)^{|X|-|S|} g(X) = \sum_{S \subseteq X \subseteq T} (-1)^{|X|-|S|} \sum_{X \subseteq Y \subseteq T} f(Y).$$

Regrouping by Y gives

$$\sum_{S \subseteq Y \subseteq T} f(Y) \sum_{S \subseteq X \subseteq Y} (-1)^{|X|-|S|}.$$

Every set X with $S \subseteq X \subseteq Y$ is obtained by adding to S some subset of $Y \setminus S$. Thus

$$\sum_{S \subseteq X \subseteq Y} (-1)^{|X|-|S|} = \sum_{A \subseteq Y \setminus S} (-1)^{|A|} = (1-1)^{|Y \setminus S|}.$$

Hence this inner sum is 1 if $Y = S$ and 0 otherwise. Therefore only the term $Y = S$ remains, so the whole expression equals $f(S)$. □

2.26 — Nash-Williams' Lemma

Lemma 2.26 (Nash-Williams' Lemma)**.** *Let G be a graph, F a spanning subgraph of G, and H an edge reconstruction of G that is not isomorphic to G. Then*

$$|G \to G|_F - |G \to H|_F = (-1)^{e(G)-e(F)} \operatorname{aut}(G). \tag{2.8}$$

Proof. Here $|G \to H|_X$ denotes the number of maps from $V(G)$ to $V(H)$ that are isomorphisms from X to a subgraph of H, and $\operatorname{aut}(G)$ is the number of automorphisms of G.

For any spanning subgraph X of G,

$$\sum_{F \subseteq X \subseteq G} |G \to H|_X = \operatorname{aut}(F) \binom{H}{F}.$$

Identifying each spanning subgraph with its edge set and applying the Möbius Inversion Formula (Theorem 2.25) gives

$$|G \to H|_F = \sum_{F \subseteq X \subseteq G} (-1)^{e(X)-e(F)} \operatorname{aut}(X) \binom{H}{X}.$$

Now compare this formula for H and for G. Since H is an edge reconstruction of G, every proper spanning subgraph X of G satisfies

$$\binom{H}{X} = \binom{G}{X},$$

so all proper-subgraph terms cancel in the difference

$$|G \to G|_F - |G \to H|_F.$$

Only the term $X = G$ remains. Since $\binom{G}{G} = 1$ and $\binom{H}{G} = 0$ (because $H \ncong G$), we obtain

$$|G \to G|_F - |G \to H|_F = (-1)^{e(G)-e(F)} \operatorname{aut}(G).$$

□

2.27 — Two Sufficient Conditions for Edge Reconstructibility

Theorem 2.27. *A graph G is edge reconstructible if there exists a spanning subgraph F of G such that either of the following two conditions holds.*

(i) $|G \to H|_F$ *takes the same value for all edge reconstructions H of G,*

(ii) $|F \to G| < 2^{e(G)-e(F)-1} \operatorname{aut}(G)$.

Proof. Let H be an edge reconstruction of G.

If condition (i) holds, then the left-hand side of (2.8) is zero, while the right-hand side is nonzero. This contradicts Nash-Williams' Lemma (Lemma 2.26).

Now suppose condition (ii) holds. By identity (2.6), the inequality

$$|F \to G| < 2^{e(G)-e(F)-1} \operatorname{aut}(G)$$

is equivalent to

$$\sum_{F \subseteq X \subseteq G} |G \to G|_X < 2^{e(G)-e(F)-1} \operatorname{aut}(G).$$

Among the spanning subgraphs X with $F \subseteq X \subseteq G$, exactly half satisfy that $e(G) - e(X)$ is even. Therefore the above inequality implies that

$$|G \to G|_X < \operatorname{aut}(G)$$

for some spanning subgraph X of G such that $e(G) - e(X)$ is even.

Applying Nash-Williams' Lemma with $F := X$, we get

$$|G \to G|_X - |G \to H|_X = (-1)^{e(G)-e(X)} \operatorname{aut}(G) = \operatorname{aut}(G),$$

since $e(G) - e(X)$ is even. Hence

$$|G \to H|_X = |G \to G|_X - \operatorname{aut}(G) < 0,$$

which is impossible.

Thus in either case $H \cong G$. Therefore G is edge reconstructible. □

2.28 — Numerical Conditions for Edge Reconstructibility

Corollary 2.28. *A graph G is edge reconstructible if either $m > \frac{1}{2}\binom{n}{2}$ or $2^{m-1} > n!$*

Proof. (Here $n = v(G)$ and $m = e(G)$.)

We use Theorem 2.27 with the choice F being the *empty* spanning subgraph of G (i.e. the graph on $V(G)$ with no edges), so $e(F) = 0$.

Case 1: $m > \frac{1}{2}\binom{n}{2}$. With F empty, $|G \to H|_F$ counts the number of bijections $V(G) \to V(H)$ that preserve the empty graph—which is simply $n!$ for any H on n vertices. Since this value is the same for all edge reconstructions H, condition (i) of Theorem 2.27 is satisfied.

Case 2: $2^{m-1} > n!$. Again with F empty, $|F \to G| = n!$ (any bijection preserves the empty graph). Condition (ii) of Theorem 2.27 requires $|F \to G| < 2^{e(G)-e(F)-1}\,\mathrm{aut}(G)$, i.e. $n! < 2^{m-1}\,\mathrm{aut}(G)$. Since $\mathrm{aut}(G) \geq 1$, the hypothesis $2^{m-1} > n!$ guarantees this.

In either case, Theorem 2.27 implies G is edge reconstructible. $\square$

3 Connected Graphs, Euler Tours, and the Friendship Theorem

3.1 — The Friendship Theorem

Theorem 3.1 (Friendship Theorem)**.** *Let G be a simple graph in which any two vertices have exactly one common neighbour. Then G has a vertex of degree $n-1$.*

Intuition. If every pair of people has exactly one mutual friend, then there must be someone who is friends with everyone (a "politician"). The proof assumes no such vertex exists, deduces the graph must be perfectly regular, and uses the eigenvalue structure of $A^2 = J + (k-1)I$ to derive that $k = 2$ and $n = 3$, which contradicts having no universal vertex.

Proof Sketch. Assume for contradiction that $\Delta < n-1$. Show that G must be k-regular for some $k \geq 2$ by proving non-adjacent vertices share a degree via a neighbourhood bijection, and then showing adjacent vertices must also share that degree. Count 2-paths: $n\binom{k}{2} = \binom{n}{2}$, giving $n = k^2 - k + 1$. Use the adjacency matrix: $A^2 = J + (k-1)I$ (because $(A^2)_{uv} = \sum A_{uw}A_{wv}$ strictly counts mutual friends). From eigenvalues of J, deduce A has eigenvalues k (multiplicity 1) and $\pm\sqrt{k-1}$ (total multiplicity $n-1$). Since $\mathrm{tr}(A) = 0$, we need $t\sqrt{k-1} = k$ for some integer t. This forces $k = 2$, $n = 3$ (just a triangle), contradicting $\Delta < n-1$.

Step-by-step proof.

1. Suppose for contradiction that no vertex has degree $n-1$ (i.e., $\Delta < n-1$).

2. G **is regular:** First, pick two non-adjacent vertices u and v. They share exactly one common neighbour w. For any other neighbour $x \in N(u)$, x cannot be adjacent to v. Because $x \nsim v$, they must share exactly one common neighbour $y \in N(v)$. This creates a perfect 1-to-1 bijection between $N(u)$ and $N(v)$, meaning $d(u) = d(v)$.

3. Next, pick two adjacent vertices $u \sim v$. They share exactly one common neighbour w. Because $\Delta < n-1$, there is some vertex y not connected to u. By our previous step, $d(u) = d(y)$. If $y \nsim v$, then $d(v) = d(y) = d(u)$. If $y \sim v$, then v is the unique common neighbour of u and y. This means y cannot be connected to w (or else u, y would share both v and w). Since $y \nsim w$, $d(y) = d(w)$. Applying symmetric logic to a vertex x connected to u but not v yields $d(v) = d(x) = d(w)$. Therefore $d(u) = d(w) = d(v)$. Thus, G is k-regular for some $k \geq 2$.

4. **Count 2-paths:** Each vertex is the centre of $\binom{k}{2}$ paths of length 2, and each pair of vertices determines exactly one such path (their unique common neighbour). So $n\binom{k}{2} = \binom{n}{2}$, giving $n = k^2 - k + 1$.

5. **Eigenvalue argument:** By the definition of matrix multiplication, the entry in row u and column v of A^2 is $(A^2)_{uv} = \sum_{w=1}^{n} A_{uw}A_{wv}$. Because A only contains 1s and 0s, a term $A_{uw}A_{wv} = 1$ if and only if vertex w is adjacent to both u and v. Thus, this sum exactly counts the number of common neighbours of u and v.

6. By the friendship condition, this count is 1 for all distinct pairs ($u \neq v$). For the diagonal ($u = v$), $(A^2)_{uu} = \sum_w A_{uw}^2$, which simply sums the 1s for every neighbour of u, giving its degree k. Therefore, $A^2 = J + (k-1)I$, where J is the all-ones matrix and I the identity matrix.

7. The eigenvalues of J are n (multiplicity 1) and 0 (multiplicity $n-1$). So eigenvalues of A^2 are $n + k - 1 = k^2$ (multiplicity 1) and $k-1$ (multiplicity $n-1$).

8. Thus A has eigenvalue k (multiplicity 1) and eigenvalues $\pm\sqrt{k-1}$ with total multiplicity $n-1$.

9. Since G is simple, $\mathrm{tr}(A) = 0$. Writing the eigenvalue sum: $k + a\sqrt{k-1} + b(-\sqrt{k-1}) = 0$ where $a + b = n - 1$. So $(a-b)\sqrt{k-1} = -k$, i.e., $t\sqrt{k-1} = k$ for integer $t = b - a$.

10. Squaring: $t^2(k-1) = k^2$, so $k^2 - t^2k + t^2 = 0$. For integer t: $k = \frac{t^2 \pm \sqrt{t^4 - 4t^2}}{2} = \frac{t^2 \pm t\sqrt{t^2-4}}{2}$. For this to be a positive integer, $t^2 - 4$ must be a perfect square. This only works for $t = \pm 2$, giving $k = 2$ and $n = 3$.

11. But K_3 (a triangle) has $\Delta = 2 = n - 1$, contradicting our assumption.

Proof to memorize. Suppose for contradiction that $\Delta < n - 1$ (no vertex is adjacent to all others).

Step 1: G is connected with $\delta \geq 2$. First, $\delta \geq 2$: for any vertex v, pick two other vertices u_1, u_2; their common neighbours with v give v at least two neighbours (even if the common neighbour is the same for both, the friendship condition applied to v and each u_i ensures v has neighbours). If two vertices were in different components, they would have no common neighbour. So G is connected.

Step 2: G is k-regular. First, we show that any two non-adjacent vertices u, v have the same degree. Since $u \nsim v$, they share exactly one common neighbour, w. Let x be any other neighbour of u ($x \neq w$). Since u and v share *only* w, x cannot be adjacent to v. Because $x \nsim v$, they must share exactly one common neighbour, say y. Since $y \sim v$, we know $y \in N(v)$. This logical relationship maps every unique neighbour in $N(u)$ to a unique neighbour in $N(v)$. Thus, there is a perfect bijection between $N(u)$ and $N(v)$, meaning $d(u) = d(v)$.

Second, we show that any two adjacent vertices $u \sim v$ also have the same degree. They share exactly one common neighbour w. Because $\Delta < n - 1$, there exists some vertex y not connected to u. By our bijection rule above, $d(u) = d(y)$.

- If y is also not connected to v, then $d(v) = d(y) = d(u)$, and their degrees match.
- If y *is* connected to v, then y and u are non-adjacent but share v as a neighbour. Since they can only share exactly one neighbour, v is their unique mutual friend. This means y cannot be connected to w. Since $y \nsim w$, our bijection rule dictates $d(y) = d(w)$. Applying this exact symmetric logic to some vertex x that is connected to u but not v reveals $d(x) = d(w)$.

Therefore, $d(u) = d(y) = d(w) = d(x) = d(v)$. Since non-adjacent and adjacent vertices share the same degree, G is perfectly k-regular with $k \geq 2$.

Step 3: $n = k^2 - k + 1$. Count paths of length 2 in two ways. Each vertex is the centre of $\binom{k}{2}$ such paths (choose 2 of its k neighbours), giving $n\binom{k}{2}$ total. Each unordered pair of vertices determines exactly one 2-path (through their unique common neighbour), giving $\binom{n}{2}$ total. So:

$$n \cdot \frac{k(k-1)}{2} = \frac{n(n-1)}{2} \implies k(k-1) = n - 1 \implies n = k^2 - k + 1.$$

Step 4: Eigenvalue analysis via $A^2 = J + (k-1)I$. Let A be the adjacency matrix. By the definition of matrix multiplication, the entry at row u and column v of A^2 is the dot product:

$$(A^2)_{uv} = \sum_{w=1}^{n} A_{uw}A_{wv}$$

Because the entries are strictly 1s and 0s, the product $A_{uw}A_{wv} = 1$ if and only if vertex w is connected to both u and v. Thus, the sum exactly counts the mutual friends of u and v.

By the Friendship Theorem's core rule, every pair of distinct vertices has exactly one mutual friend, so $(A^2)_{uv} = 1$ for all $u \neq v$. On the diagonal where $u = v$, the sum $(A^2)_{uu} = \sum_w A_{uw}^2$ simply counts every friend of u, which is its degree k. Therefore, A^2 has k's on the diagonal and 1's everywhere else:

$$A^2 = J + (k-1)I,$$

where J is the all-ones matrix and I the identity.

The eigenvalues of J are n (multiplicity 1, eigenvector $\mathbf{1}$) and 0 (multiplicity $n-1$). So $A^2 = J + (k-1)I$ has eigenvalues:

$$n + (k-1) = k^2 \quad \text{(mult. 1)}, \qquad 0 + (k-1) = k-1 \quad \text{(mult. } n-1\text{)}.$$

The eigenvalues of A are therefore $\pm k$ (from k^2) and $\pm\sqrt{k-1}$ (from $k-1$). Since G is k-regular, $A\mathbf{1} = k\mathbf{1}$, so k is an eigenvalue of A with eigenvector $\mathbf{1}$, and the eigenvalue $-k$ does not appear (any eigenvector for $-k$ would be orthogonal to $\mathbf{1}$, but one checks this leads to all-zero entries). So A has eigenvalue k with multiplicity 1 and eigenvalues $+\sqrt{k-1}$, $-\sqrt{k-1}$ with multiplicities a, b satisfying $a + b = n - 1$.

Step 5: Trace forces $k = 2$, $n = 3$. Since G is simple, $\mathrm{tr}(A) = 0$. The trace is the sum of eigenvalues:

$$k + a\sqrt{k-1} - b\sqrt{k-1} = 0 \implies (a-b)\sqrt{k-1} = -k.$$

Set $t = b - a \in \mathbb{Z}$. Then $t\sqrt{k-1} = k$. Squaring: $t^2(k-1) = k^2$, so $k^2 - t^2k + t^2 = 0$. By the quadratic formula:

$$k = \frac{t^2 \pm \sqrt{t^4 - 4t^2}}{2} = \frac{t^2 \pm t\sqrt{t^2-4}}{2}.$$

For k to be a positive integer, $t^2 - 4$ must be a non-negative perfect square. Write $t^2 - 4 = s^2$, i.e., $(t-s)(t+s) = 4$. The only positive integer solution is $t = 2, s = 0$, giving $k = 4/2 = 2$ and $n = 4 - 2 + 1 = 3$.

But K_3 has $\Delta = 2 = n - 1$, contradicting our assumption that $\Delta < n - 1$. $\square$

3.5 — Eulerian if and only if Even

Theorem 3.5. *A connected graph is eulerian if and only if it is even.*

Proof. (⇒) Suppose G has an Euler tour. Every time the tour enters a vertex, it must also leave it, so the edges incident with that vertex are used in pairs. Hence every vertex has even degree, and G is even.

(⇐) Suppose G is connected and even. We prove that G has an Euler tour by induction on $e(G)$. Suppose every connected even graph with fewer than $e(G)$ edges has an Euler tour.

If $e(G) = 0$, then G consists of a single vertex, so the result is trivial.

Now assume $e(G) > 0$. Since G is connected and every vertex has even degree, G contains a cycle C (by Theorem 2.1). Remove the edges of C to obtain $G' = G - E(C)$. Every vertex of G' still has even degree, so each nontrivial component of G' is connected and even. Since each such component has fewer edges than G, by induction it has an Euler tour.

Finally, traverse the cycle C, and whenever a vertex of C lies in a nontrivial component of G', detour through an Euler tour of that component and then return to the same vertex of C. In this way we obtain a closed walk using every edge of G exactly once. Thus G is eulerian. □

3.8 — From a Double Cover-Free Covering to a Cycle Double Cover

Proposition 3.8. *If a graph has a cycle covering in which each edge is covered at most twice, then it has a cycle double cover.*

Proof. Let $\mathscr{F}$ be a cycle covering of G in which each edge is covered at most twice. Let E_1 be the set of edges covered exactly once. Consider the symmetric difference

$$H = \triangle\{E(C) : C \in \mathscr{F}\}.$$

An edge covered once appears in H, and an edge covered twice does not. Hence $H = E_1$. Each cycle is an even subgraph, and the symmetric difference of even subgraphs is again even (Corollary 2.16). Therefore H is an even subgraph. By Veblen's Theorem (Theorem 2.7), H has a cycle decomposition $\mathscr{F}'$.

Now every edge in E_1 is covered once by $\mathscr{F}$ and once by $\mathscr{F}'$, while every other edge was already covered twice by $\mathscr{F}$ and does not belong to $\mathscr{F}'$. Hence $\mathscr{F} \cup \mathscr{F}'$ is a cycle double cover of G. $\square$

4 Trees

4.1 — Unique Paths in a Tree

Proposition 4.1. *In a tree, any two vertices are connected by exactly one path.*

Proof. Since a tree is connected, any two vertices are joined by at least one path.

Suppose there were two distinct paths between vertices u and v. Then their union would contain a cycle, contradicting that a tree is acyclic. Therefore there is exactly one path between any two vertices. □

4.2 — Every Nontrivial Tree has Two Leaves

Proposition 4.2. *Every nontrivial tree has at least two leaves.*

Proof. Let $P = v_1 v_2 \cdots v_k$ be a longest path in T.

If v_1 were not a leaf, it would have a neighbour $u \neq v_2$. This vertex u cannot lie on P, for otherwise T would contain a cycle. But then

$$u, v_1, v_2, \ldots, v_k$$

would be a path longer than P, a contradiction. Thus v_1 is a leaf. Similarly, v_k is a leaf.

Hence every nontrivial tree has at least two leaves. □

4.3 — Trees have $v - 1$ Edges

Theorem 4.3. *If T is a tree, then $e(T) = v(T) - 1$.*

Proof. We use induction on $v(T)$.

If $v(T) = 1$, then $e(T) = 0 = v(T) - 1$.

Now let $v(T) \geq 2$, and assume the result holds for all trees with fewer vertices. By Proposition 4.2, T has a leaf v. Then $T - v$ is again a tree, with one fewer vertex and one fewer edge. By the induction hypothesis,

$$e(T - v) = v(T - v) - 1 = v(T) - 2.$$

Therefore

$$e(T) = e(T - v) + 1 = v(T) - 1.$$

□

4.5 — Branchings in Tournaments

Theorem 4.5. *Any tournament on $2k$ vertices contains a copy of each branching on $k + 1$ vertices.*

Proof. Let $v_1, \ldots, v_{2k}$ be a median order of T. We prove the statement by induction on k.

If $k = 1$, then B has two vertices and one arc, which every tournament on 2 vertices contains.

Now let $k \geq 2$. Choose a leaf y of B, and let x be its predecessor. Put $B' := B - y$, so B' is a branching on k vertices. Also let

$$T' := T - \{v_{2k-1}, v_{2k}\}.$$

Then T' is a tournament on $2(k-1)$ vertices, and $v_1, \ldots, v_{2k-2}$ is a median order of T'. By induction, T' contains a copy of B'.

Let v_i be the vertex of this copy corresponding to x. By the median-order property, v_i dominates at least half of the vertices to its right. Since the copy of B' uses only k vertices altogether, not all vertices to the right of v_i can be both used and dominated by v_i. Hence there is a vertex v_j to the right of v_i which is not used by the copy of B' and satisfies $v_i v_j \in E(T)$.

Place y at v_j. Then the arc joining x to y in B is realized by the arc $v_i v_j$ in T, so this extends the copy of B' to a copy of B.

Therefore T contains a copy of B.

□

4.6 — Connected if and only if there is a Spanning Tree

Theorem 4.6. *A graph is connected if and only if it has a spanning tree.*

Proof. ($\Leftarrow$) If G has a spanning tree, then that tree is connected and contains all vertices of G, so G is connected.

($\Rightarrow$) Suppose G is connected. If G is already acyclic, then it is a spanning tree. Otherwise, remove an edge from a cycle. This does not disconnect the graph. Repeating this process finitely many times yields a connected acyclic spanning subgraph of G, that is, a spanning tree. □

4.7 — Bipartite if and only if No Odd Cycle

Theorem 4.7. *A graph is bipartite if and only if it contains no odd cycle.*

Proof. It is enough to prove the result for connected graphs.

($\Rightarrow$) If G is bipartite with bipartition (X, Y), then every walk alternates between X and Y. In particular, every cycle has even length. So G contains no odd cycle.

($\Leftarrow$) Suppose G is connected and contains no odd cycle. Let T be a spanning tree of G (Theorem 4.6), and fix a root r. Define

$$X = \{v : d_T(r, v) \text{ is even}\}, \qquad Y = \{v : d_T(r, v) \text{ is odd}\},$$

where $d_T(r, v)$ denotes the distance from r to v in T.

We show that every edge of G joins a vertex of X to a vertex of Y. If $uv \in E(T)$, then

$$|d_T(r, u) - d_T(r, v)| = 1,$$

so u and v have opposite parity.

Now let $uv \in E(G) \setminus E(T)$. If u and v had the same parity, then the unique u-v path in T would have even length. Adding the edge uv would then create an odd cycle, contrary to hypothesis. So again u and v have opposite parity.

Thus every edge joins X to Y, and hence (X, Y) is a bipartition of G. □

4.8 — Cayley's Formula

Theorem 4.8 (Cayley's Formula)**.** *The number of labelled trees on n vertices is n^{n-2}.*

Proof. **Claim:** The number of labelled *branchings* (rooted trees) on n vertices is n^{n-1}.

Construction. Build a branching by adding directed edges (u, v) one at a time to n isolated vertices, where u, v lie in different components and v is the root of its component (so edges point toward roots). Each edge merges two components, so after $n-1$ edges we have a single rooted tree.

Counting choices. When k components remain: u can be any of the n vertices, and v must be the root of one of the other $k-1$ components. So there are $n(k-1)$ choices. As k runs through $n, n-1, \ldots, 2$, the total number of construction sequences is

$$\prod_{k=2}^{n} n(k-1) \;=\; n^{n-1}(n-1)!.$$

Correcting overcount. Each branching has $n-1$ edges, and any of the $(n-1)!$ orderings of its edges yields a valid construction sequence. So each branching is counted exactly $(n-1)!$ times, giving

$$(\text{number of branchings}) \;=\; \frac{n^{n-1}(n-1)!}{(n-1)!} \;=\; n^{n-1}.$$

From branchings to trees. Each labelled tree yields exactly n branchings (one per choice of root; orientations are then forced). Hence

$$(\text{number of labelled trees}) \;=\; \frac{n^{n-1}}{n} \;=\; n^{n-2}. \qquad \square$$

4.10 — Even Subgraphs from Fundamental Cycles

Theorem 4.10. *Let T be a spanning tree of a connected graph G, and let S be a subset of its cotree $\overline{T}$. Then $C := \triangle\{C_e : e \in S\}$ is an even subgraph of G. Moreover, $C \cap \overline{T} = S$, and C is the only even subgraph of G with this property.*

Proof. For each edge $e \in \overline{T}$ (i.e. $e \in E(G) \setminus E(T)$), the *fundamental cycle* C_e is the unique cycle in $T + e$. It is an even subgraph (every vertex has degree 0 or 2).

C **is even:** The symmetric difference of even subgraphs is even (since adding degrees mod 2 preserves evenness; see Corollary 2.16). So $C = \triangle\{C_e : e \in S\}$ is even.

$C \cap \overline{T} = S$**:** Each fundamental cycle C_e contains exactly one cotree edge, namely e itself (all other edges of C_e are in T). So when we take the symmetric difference over $e \in S$, a cotree edge e appears in exactly one of the C_e's, so $e \in C$. A cotree edge $e' \notin S$ appears in none of the C_e's, so $e' \notin C$. Hence $C \cap \overline{T} = S$.

Uniqueness: Suppose C' is another even subgraph with $C' \cap \overline{T} = S$. Then $C \triangle C'$ is an even subgraph (symmetric difference of two even subgraphs) with $(C \triangle C') \cap \overline{T} = S \triangle S = \emptyset$. So $C \triangle C'$ is contained entirely in T. But the only even subgraph of a tree is the empty graph (a tree has no cycles, and any nonempty even subgraph contains a cycle). Therefore $C \triangle C' = \emptyset$, i.e. $C' = C$. ☐

4.11 — Unique Expression as a Symmetric Difference

Corollary 4.11. *Let T be a spanning tree of a connected graph G. Every even subgraph of G can be expressed uniquely as a symmetric difference of fundamental cycles with respect to T.*

Proof. Let C be an even subgraph of G, and set

$$S := C \cap \overline{T}.$$

By Theorem 4.10, the symmetric difference

$$\triangle\{C_e : e \in S\}$$

is the unique even subgraph whose cotree edges are exactly S. Since C is also an even subgraph with cotree edges exactly S, it follows that

$$C = \triangle\{C_e : e \in S\}.$$

Uniqueness of the representation is immediate from the uniqueness part of Theorem 4.10. ☐

4.12 — Every Cotree Lies in a Unique Even Subgraph

Corollary 4.12. *Every cotree of a connected graph is contained in a unique even subgraph of the graph.*

Proof. Let T be a spanning tree of a connected graph G, and let $\overline{T}$ be its cotree. Apply Theorem 4.10 with

$$S = \overline{T}.$$

Then there is a unique even subgraph C of G such that

$$C \cap \overline{T} = \overline{T}.$$

Thus C contains every cotree edge, and this even subgraph is unique. $\square$

5 Nonseparable Graphs

5.1 — No Cut Vertices and Internally Disjoint Paths

Theorem 5.1. *A connected graph on three or more vertices has no cut vertices if and only if any two distinct vertices are connected by two internally disjoint paths.*

Proof. ($\Leftarrow$) Suppose any two distinct vertices of G are connected by two internally disjoint paths. Let v be any vertex of G, and let $a, b \in V(G) \setminus \{v\}$ be distinct. Since a and b are joined by two internally disjoint paths, at most one of these paths can pass through v. Hence at least one avoids v, so a and b are connected in $G - v$. Therefore $G - v$ is connected, and so v is not a cut vertex. Since v was arbitrary, G has no cut vertices.

($\Rightarrow$) Suppose G is connected and has no cut vertices. Let u, v be distinct vertices of G. Choose a shortest u-v path

$$P : u = v_0, v_1, \ldots, v_k = v.$$

If $k = 1$, then $uv \in E(G)$. Since u is not a cut vertex, $G - u$ is connected, so in $G - u$ there is a path from v to some neighbour of u. Let w be the first vertex on such a path that is adjacent to u. Then this gives a u-v path different from the edge uv. Together with the edge uv, we obtain two internally disjoint u-v paths.

Now assume $k \geq 2$, so P has an internal vertex; let $x = v_1$. Since x is not a cut vertex, the graph $G - x$ is connected. Hence in $G - x$ there is a path from u to v. Among all such paths, choose one, say Q, so that the set $V(P) \cap V(Q)$ is as small as possible.

We claim that P and Q are internally disjoint. Suppose not. Then some internal vertex y of P also lies on Q. Let y be the first such vertex along Q starting from u. Then the initial segment of Q from u to y meets P only in its ends. Replacing the u-y segment of P by this segment of Q gives a u-v path shorter than P, contradicting the choice of P as a shortest u-v path.

Thus P and Q are two internally disjoint u-v paths. Since u, v were arbitrary, any two distinct vertices are connected by two internally disjoint paths. $\square$

6 Optimal Trees

6.10 — Every Jarník–Prim Tree is Optimal

Theorem 6.10. *Every Jarník–Prim tree is an optimal tree.*

Proof. Let T be a Jarník–Prim tree of a connected weighted graph G, rooted at r. We prove by induction on $v(T)$ that T is optimal.

Base case: If $v(T) = 1$, then T has no edges, so it is trivially optimal.

Inductive step: Assume $v(T) \geq 2$, and let e be the first edge chosen by the algorithm. Thus e has minimum weight among all edges incident to r.

We first show that some optimal spanning tree contains e. Let T^* be an optimal spanning tree. If $e \in E(T^*)$, there is nothing to prove. Otherwise, $T^* + e$ contains a unique cycle C. Let f be the other edge of C incident to r. Then

$$T^{**} := (T^* + e) \setminus \{f\}$$

is a spanning tree, and since $w(e) \leq w(f)$,

$$w(T^{**}) = w(T^*) + w(e) - w(f) \leq w(T^*).$$

By optimality of T^*, T^{**} is also optimal. Hence some optimal spanning tree contains e.

Now contract e to obtain $G' := G/e$, let r' be the contracted vertex, and let $T' := T/e$. Spanning trees of G that contain e correspond naturally to spanning trees of G', and optimal trees containing e correspond to optimal trees of G'.

Also, T' is a Jarník–Prim tree of G' rooted at r', since after choosing e, the remaining steps of Prim's algorithm on G are exactly the steps of Prim's algorithm on G'.

Because G' has fewer vertices, the induction hypothesis implies that T' is optimal in G'. Therefore T is optimal in G. □